AUTOMOBILE ENGINEERING · PART 2 OF 12

The Engine, Part 2: Power, Torque and Mileage

By Isagi · 9 November 2026 · All specs as of November 2026 (manufacturer brochures and listings)

TL;DR

Power is how fast an engine can do work; torque is how hard it can push. You feel torque when pulling away or climbing, and power when overtaking at speed. To pick a car, look at the **spread** of torque (the rpm range where it is available) and the car's power-to-weight, not the single biggest number. Mileage is decided by how much energy it takes to move the car (weight, drag, tyres) and how efficiently the engine and gearbox turn fuel into that energy. ARAI figures are lab numbers; owners of the Creta variants report city figures roughly 17% to 46% lower.

1. Torque and power in plain words

Torque is the twisting force the engine puts on the crankshaft, measured in newton-metres (Nm). **Power** is the rate of doing work, which is torque multiplied by how fast the engine is turning. The conversion is worth knowing once:

Power (kW) = Torque (Nm) × rpm ÷ 9,549, and **bhp = kW ÷ 0.746**.

Use it on the Swift's spec sheet. Peak torque is 111.7 Nm at 4,300 rpm, which works out to about 50 kW or 67 bhp at that moment. The 80.46 bhp peak arrives later, at 5,700 rpm, where torque has slipped to about 100 Nm (60 kW × 9,549 ÷ 5,700). Torque falls at high rpm because the engine has less time to fill its cylinders, but rpm keeps rising, so power keeps climbing for a while longer.

Illustrative curves: the shape is typical; only the marked points come from the spec sheets

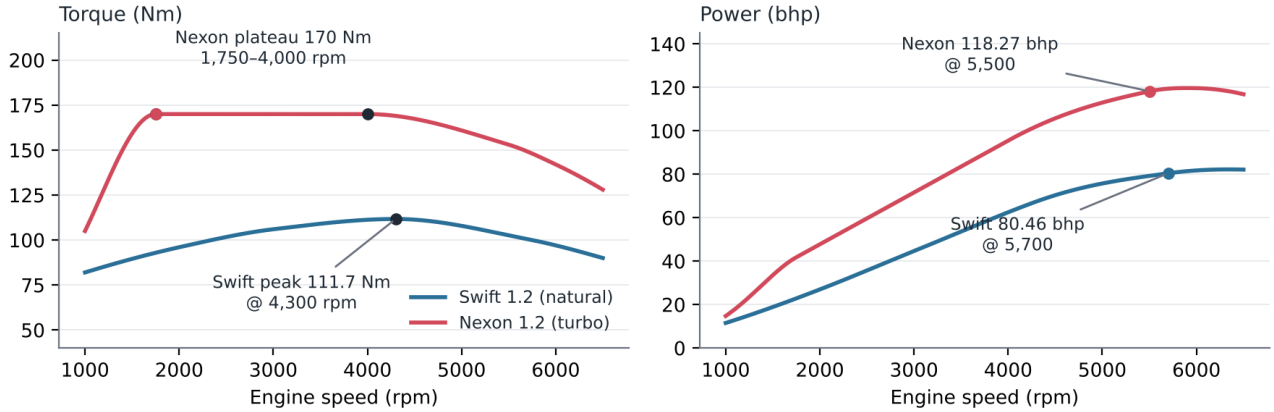


Fig 1. Torque (left) and power (right) against engine speed. The Nexon's turbo gives it a flat torque plateau from low rpm; the naturally aspirated Swift peaks later and lower.

2. Reading the numbers on a spec sheet

A spec line such as **118.27 bhp @ 5,500 rpm, 170 Nm @ 1,750–4,000 rpm** tells you three things: how much, where it peaks, and how wide the torque plateau is. The plateau matters most in daily driving.

- **Nexon 1.2 turbo:** 170 Nm is available from just 1,750 rpm to 4,000 rpm. The car pulls without you changing gear.

- **Creta 1.5 diesel:** 250 Nm from 1,500 to 2,750 rpm. At 2,750 rpm that is about 72 kW or 96 bhp, strong pull from barely above idle, which is why diesels feel effortless in traffic and on climbs.
- **Creta 1.5 MPi petrol:** 143.8 Nm only at 4,500 rpm. You need to keep revs up, so expect more downshifting.

Torque at the wheels is engine torque multiplied by the gear ratio, so gearing changes how a car feels as much as the engine does. Part 3 covers that.

3. How to pick a car by power

- **Start with your driving.** City crawl, mixed use, regular highway, hills, a full family load: each rewards a different number.
- **Look at the torque spread,** not just the peak. A wide plateau starting below 2,000 rpm means relaxed driving.
- **Check power-to-weight:** bhp divided by kerb weight in tonnes. The Swift's 80.46 bhp over a listed 925 kg is about 87 bhp per tonne.* As a rough guide, under 70 feels leisurely, 80 to 100 is brisk enough for most buyers, and above 120 is quick.*
- **Remember the gearbox.** An automatic or CVT can dull a small engine's response; a turbo petrol with a DCT feels sharper.
- **Don't pay for power you can't use.** Speed limits and traffic cap how much extra power you will ever use.

If you mostly...	Look for	Examples from our list
Drive in the city, 1–2 people	Light weight, easy low-rpm response, an easy automatic	Swift 1.2; Nexon 1.2 turbo (plateau from 1,750 rpm)
Drive mixed with a family of four	Torque available from about 1,500–2,000 rpm	Nexon 1.2 turbo; Creta 1.5 diesel
Do highway runs and overtake often	Power at speed, good power-to-weight, a tall top gear	Creta 1.5 turbo (158 bhp); Skoda 1.5 TSI (150 PS)
Climb hills or carry full loads	A wide torque plateau; diesel or turbo	Creta 1.5 diesel; Thar diesel (about 300 Nm)

4. What actually makes mileage better

Mileage depends on two things: how much energy the car needs, and how well the engine and drivetrain supply it.

Energy the car needs. Three forces resist motion. *Weight* matters whenever you accelerate or climb: more mass, more fuel. *Rolling resistance* comes from the tyres and rises with weight and low tyre pressure. *Aerodynamic drag* grows with the square of speed, so going from 100 to 120 km/h raises drag force by 44% and the power needed to beat it by about 73%. Speed is the biggest lever a driver controls.

How well the engine converts fuel. Textbooks put the best-case thermal efficiency of a petrol engine at roughly 30–35% and a diesel at roughly 40%; the rest leaves as heat.* Real driving is worse, because city driving uses only a small fraction of the engine's capability, where friction and pumping losses take a bigger share. That is why start-stop, cylinder deactivation and hybrids help most in the city.

Drivetrain and conditions. More gear ratios keep the engine in its efficient range. Air conditioning, traffic, tyre pressure, a loaded boot and a heavy foot all change the result.

5. ARAI versus real-world mileage

The ARAI figure comes from a standard test on a laboratory roller, with fixed speeds, no air conditioning and no traffic. It is good for comparing cars with each other, not for predicting your fuel bill. The Hyundai Creta shows the gap clearly:

Creta 1.5 variant	ARAI claim (km/l)	City, owner-reported	Highway, owner-reported	City shortfall
Petrol, manual	17.4	12–14	up to about 17	20–31%
Turbo petrol, DCT	18.4	10–12	16–17	35–46%
Diesel, manual	21.8	16–18	over 20	17–27%

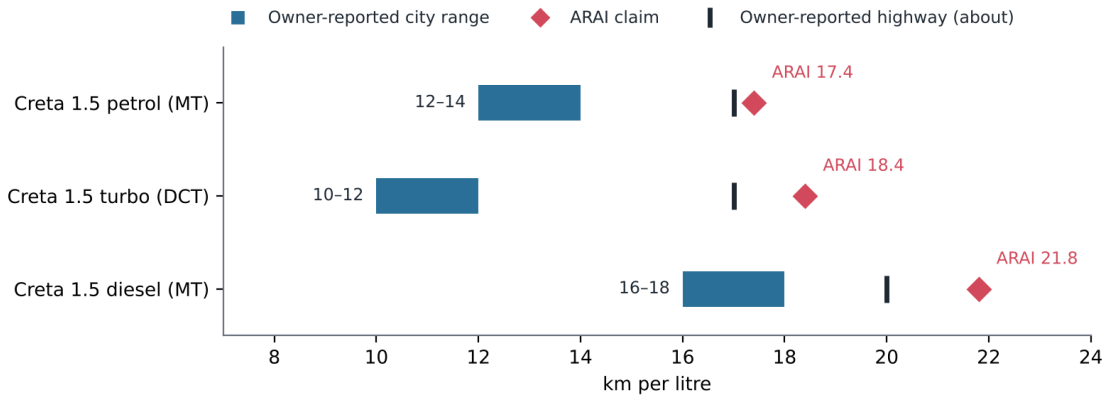


Fig 2. ARAI claim versus owner-reported figures for three Creta 1.5 variants. Owner-reported figures vary with traffic, AC use and driving style.

Note the turbo petrol: it has the best ARAI figure of the three petrols and the worst city figure. Boost helps in a gentle lab test and costs fuel when real traffic makes you use it.

6. Turbo, downsizing and hybrids

Downsizing replaces a big engine with a small turbocharged one (Section 8 of Part 1: Skoda's 1.0 TSI, Tata's 1.2 Revotron). A small engine at steady cruise runs at a higher load, which is more efficient, and the turbo supplies extra pull on demand. The weakness is the one in Fig 2: when you demand the boost, the engine burns more fuel.

Cylinder deactivation shuts some cylinders at light load; Skoda's 1.5 TSI is reported to do this on two of its four.* **Diesel** is more efficient than petrol at the same size, but costs more to buy and to keep BS6-compliant.

Hybrids change the picture most. The Honda City e:HEV claims about 27.3 km/l and the Toyota Hyryder strong hybrid about 28.0 km/l (ARAI), roughly 57–61% more than the petrol Creta's 17.4.* These are different cars, so the comparison shows scale rather than an exact gain. Part 7 covers hybrids properly.

7. BS6 and E20: what changed for engines

India moved to **BS6** emission norms on 1 April 2020, and to a stricter second phase on 1 April 2023 that added real-driving emission testing and on-board diagnostics. For petrol engines this mostly meant better catalytic converters and engine tuning. For diesels it meant a particulate filter and a urea-based SCR system, which add cost, and some makers, notably Maruti Suzuki, stopped selling small diesels altogether.*

E20 petrol (20% ethanol) is now the standard pump fuel nationwide. Ethanol carries less energy per litre than petrol. The petroleum ministry's estimate (August 2025) is a mileage loss of about

1–2% for E10-designed cars calibrated for E20 and 3–6% for others; the transport minister told Parliament in July 2026 that the drop is 2–6%.

Summary

- Torque is push; power is push times speed. kW = Nm × rpm ÷ 9,549, and bhp = kW ÷ 0.746.
- Peak torque comes at lower rpm than peak power. The width of the torque plateau decides how relaxed a car feels.
- To pick a car, match the torque spread and power-to-weight to your driving. Do not chase the biggest number.
- Mileage = energy the car needs (weight, tyres, drag that grows with speed squared) ÷ how efficiently the engine and drivetrain deliver it.
- ARAI numbers compare cars; real-world city figures for the Creta variants run 17–46% lower.
- Small turbos help at cruise but cost fuel when you use the boost. Hybrids give the biggest real gain.
- BS6 made diesels costlier; E20 may trim km/l slightly for older petrols.

*Unverified or approximate: Swift kerb weight (925 kg) from a listing; power-to-weight bands, thermal-efficiency percentages, Skoda cylinder deactivation and small-diesel withdrawal are textbook rules of thumb or general knowledge, not checked against an official document. Hybrid figures: Carlelo (ARAI 27.26 and 27.97 km/l). Creta ARAI claims and owner-reported mileage: Spinny's Creta mileage comparison (owner-reported, indicative only). Spec sources as in Part 1. Theory: Heywood, Internal Combustion Engine Fundamentals. BS6 phase dates (1 April 2020 and 1 April 2023, with real-driving emissions and on-board diagnostics in Phase 2) were cross-checked in November 2026 (Autocar India, Down To Earth).

See also: Automobile Engineering 1/12 — The Engine, Part 1. Next: Part 3, Transmissions and drivetrain.